Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 25-03-2026**
Time: 09.00Am to 12.00Pm **GTM-28** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**3**	2	**2**	3	**3**	4	**1**	5	**4**
6	**2**	7	**2**	8	**2**	9	**3**	10	**1**
11	**1**	12	**2**	13	**1**	14	**4**	15	**3**
16	**2**	17	**1**	18	**3**	19	**3**	20	**1**
21	**2600**	22	**48**	23	**4**	24	**28**	25	**1**

PHYSICS

26	**2**	27	**3**	28	**1**	29	**2**	30	**2**
31	**3**	32	**4**	33	**4**	34	**3**	35	**4**
36	**3**	37	**2**	38	**1**	39	**2**	40	**2**
41	**2**	42	**2**	43	**4**	44	**3**	45	**1**
46	**73**	47	**5**	48	**9**	49	**112**	50	**2**

CHEMISTRY

51	**3**	52	**2**	53	**2**	54	**4**	55	**4**
56	**3**	57	**4**	58	**2**	59	**2**	60	**3**
61	**3**	62	**3**	63	**3**	64	**4**	65	**2**
66	**3**	67	**2**	68	**1**	69	**1**	70	**4**
71	**400**	72	**5**	73	**5**	74	**5**	75	**74**

For More Join: @IITJEE_Advanced

SOLUTION

MATHEMATICS

01. $D=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 1 & 3 & \alpha\end{vmatrix}=0,\ \alpha=5$

$D_1=\begin{vmatrix}5 & 1 & 1\\ 9 & 2 & 3\\ \beta & 3 & 5\end{vmatrix}=0,\ \beta=13,\ \ \alpha^2+\beta^2=194$

02. By using expansion

$$\lim_{x\to 0}\frac{ax\left[1+x+\frac{x^2}{2!}+....\right]-b\left[x-\frac{x^2}{2}+\frac{x^3}{3}+.....\right]+cx\left[1-\frac{x}{1!}+\frac{x^2}{2!}.....\right]}{x^3\left(\frac{\sin x}{x}\right)}=2$$

$a-b+c=0,\ a+\frac{b}{2}-c=0,\frac{a}{2}-\frac{b}{3}+\frac{c}{2}=2$

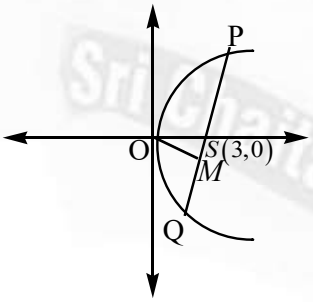
$a=3,b=12,c=9\ \ a+b+c=24$

03. $S=\sum_{r=50}^{100}\frac{{}^{100}C_1}{r+1}=\sum_{r=50}^{100}\frac{1}{r+1}\cdot\frac{r+1}{101}\cdot{}^{101}C_{r+1}$

$S=\frac{1}{101}\sum_{r=50}^{100}{}^{101}C_{r+1}=\frac{1}{101}\cdot\frac{2}{2}=\frac{2^{100}}{101}$

04. $r_1=a_1+\lambda b_1,\ r_2=a_2+\lambda b_2$ $\qquad d=\frac{(a_1-a_1)\cdot(b_1\times b_2)}{|b_1\times b_2|}=\left|\frac{-6}{\sqrt{5}}\right|=\frac{6}{\sqrt{5}}$

05. Length of focal chord $=4a\cos ec^2\theta$



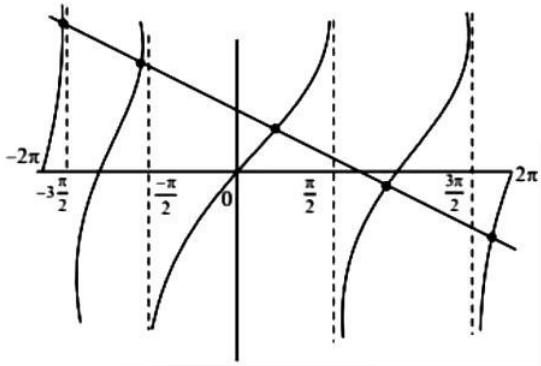
$12\cos ec^2\theta=15\qquad \sin\theta=\frac{2}{\sqrt{5}}$

From the ΔOMS, $\sin\theta=\frac{OM}{OS}=\frac{d}{3}$

$\frac{d}{3}=\frac{2}{\sqrt{5}}\Rightarrow d=\frac{6}{\sqrt{5}}\qquad 100\,d^2=100\times\frac{36}{5}=720$

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06. $Tanx = \frac{\pi}{3} - \frac{2x}{3}$



5 solutions

07. Given that $|z| \geq 3$

Now $\left|z + \frac{(6+8i)}{2z}\right| \geq |z| - \frac{|(6+8i)|}{|2z|}$

The least value occurs when $|z| = 3$ $\geq \left|3 - \frac{\sqrt{36+64}}{2\times 3}\right| = \frac{4}{3}$

08. Given quadratic equation

$x^2 - x + 2 = 0 \text{ -------} (1)$

Also given α, β are the roots of equation (1)

$S_n = 2024\alpha^n + 2025\beta^n$

Since α and β are the roots of equation (1)

$\alpha^2 - \alpha + 2 = 0$ $\qquad$ $\beta^2 - \beta + 2 = 0$

$\alpha^2 = \alpha - 2$ $\qquad$ $\beta^2 = \beta - 2$

$\alpha^{n-2}.\alpha^2 = \alpha^{n-2}.(\alpha - 2)$ $\qquad$ $\beta^n = \beta^{n-1} - 2\beta^{n-2}$ .....(3)

$\alpha^n = \alpha^{n-2}(\alpha - 1)$

$\alpha^n = \alpha^{n-1} - 2\alpha^{n-2}$ .....(2)

$\therefore S_n = 2024\alpha^n + 2025\beta^n$

$= 2024\left(\alpha^{n-1} - 2\alpha^{n-2}\right) + 2025\left(\beta^{n-1} - 2\beta^{n-2}\right)$

From equation 2, 3

$S_n = S_{n-1} - 2S_n - 2$

On putting n = 10. We have $S_{10} = S_9 - 2S_8$

09. $f(x) = \frac{1}{\sin x + \cos x - 2}$ $\qquad$ $R_f = \left[\frac{1}{-\sqrt{2}-2}, \frac{1}{\sqrt{2}-2}\right]$

$a = \frac{1}{\sqrt{2}-21} = \frac{\sqrt{2}+2}{-2}$, $\qquad$ $b = \frac{\sqrt{2}-2}{2}$

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$$\alpha=\frac{a+b}{\sqrt{2}}=\frac{1}{2}\left[\frac{\sqrt{2}+2}{-2}+\frac{\sqrt{2}-2}{2}\right]=-1$$

$$\beta=\sqrt{ab}=\sqrt{\frac{\sqrt{2}+2}{-2}\times\frac{\sqrt{2}-2}{2}}=\frac{1}{\sqrt{2}} \qquad \alpha^2-\beta^2=\frac{1}{2}$$

10. $\ell^2+m^2+n^2=1$

$$AM\geq GM,\frac{\ell^2+m^2}{2}\geq \ell m,\frac{m^2+n^2}{2}\geq mn,\frac{n^2+\ell^2}{2}\geq \ell n$$

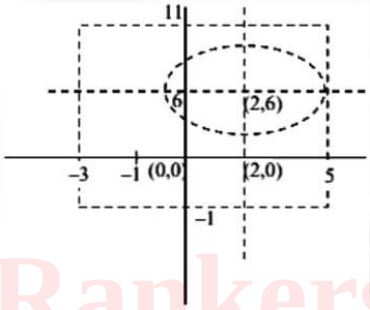
Adding $\ell^2+m^2+n^2\geq \ell m+mn+n\ell$

$\ell^2+m^2+n^2=1$

Hence $\ell m+mn+n\ell\leq 1$, maximum value 1

$\ell^2+m^2+n^2=1,\ \ell=\frac{1}{3}$,For maximum $m=n$

11. $A:|x-1|\leq 4.|y-5|\leq 6$



$\Rightarrow -4\leq x-1\leq 4\Rightarrow -3\leq x\leq 5$

$|y-5|\leq 6\Rightarrow -6\leq y\leq -5\leq 6\Rightarrow -12y\leq 1$

$$B:16(x-2)^2+9(y-6)^2\leq 144 \qquad \frac{(x-2)^2}{9}+\frac{(y-6)^2}{16}\leq 1$$

From diagram $B\subset A$

12. $$|B|=\left|2A^{-1}\right|=2^3\left|A^{-1}\right|=\frac{8}{|A|}=8$$

$$|C|=\left|\frac{adjA}{\sqrt[3]{2}}\right|=\frac{|adj\ A|}{\left(\sqrt[3]{2}\right)^3}=\frac{1}{2}|adj\ A|=\frac{|A|^2}{2}=\frac{1}{2}$$

$$\left|AB^2C^3\right|=\left|A\left(2A^{-1}\right)^2.C^3\right|=\left|4A^{-1}C^3\right|$$

$$=\left|4A^{-1}\right||C|^3=4^3\frac{1}{|A|}.\left[\frac{1}{2}\right]^3=8$$

13. Since 'C' is parallel to $a+b+c$

$(a+b+c)\times C=0\Rightarrow (a+b)\times c=0$

$$c=\mu(a+b)=2\mu\left[(\lambda+3)\bar{i}+\bar{k}\right] \qquad \ldots.(1)$$

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Now a.c $= -68, \Rightarrow 2\lambda.2\mu(\lambda+3)-2.24=-68$

$=\mu\lambda(\lambda+3)-\mu=-17$ ....(2)

And $b.c=-80\Rightarrow\lambda=\dfrac{-20-11\mu}{3\mu}$ ....(3)

From 1, 2, 3

$\Rightarrow\mu=-1,\dfrac{-40^{0}}{13},\lambda=3,\dfrac{-207}{60}$

$\lambda\in z,\lambda=3$ *and* $\mu=-1$

$c=-2(6i+k)=-12\bar{i}-2\bar{k}$

Now $c\times b=\begin{vmatrix} i & j & k \\ -12 & 0 & -1 \\ 6 & -2 & 4 \end{vmatrix}=-4\bar{i}+36\bar{j}+24\bar{k}$

$|a|=2\sqrt{11},|c\times b|=2\sqrt{118}$

$\left[\dfrac{(c\times b)^2}{|a|^2}\right]=\left[\dfrac{2\left(\sqrt{118}\right)^2}{\left(2\sqrt{11}\right)^2}\right]=10$

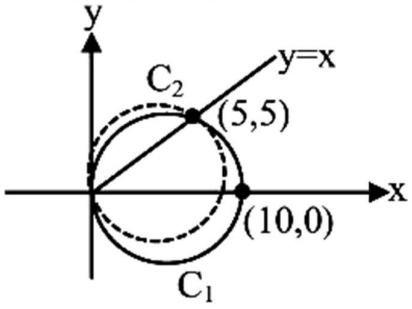
14. Equation of circle C_2 is

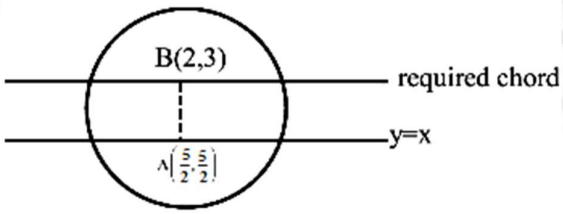
$x^2+y^2-5x-5y=0$

Its centre is $\left(\dfrac{5}{2},\dfrac{5}{2}\right)$

$m_{AB}=-1$

$\therefore$ Slope of required chord = 1





$\therefore$ Equation of required chord is $x-y+1=0$

$\therefore a=-1, b=1$

$\therefore a+b=-1+1=0$

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15. $f(x+y)=4(x).f(y)\Rightarrow f(x)=e^{kx}$

Put $x=1\Rightarrow f(1)=e^{k}\Rightarrow 2=e^{k}, f(x)=2^{x}$

Now $\sum_{r=1}^{n} f(c+r)=f(c+1)+f(c+2)+...+f(c+n)$

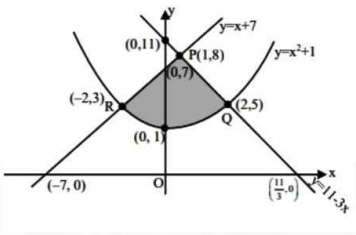
$=2^{c+1}+2^{c+2}+....+2^{c+n}$

$=2^{c}.2\left[1+2+2^{2}+...+n\,terms\right]$

$=2^{c}.2.\dfrac{1\left(2^{n}-1\right)}{2-1}=2^{c}.2\left(2^{n}-1\right)$

$\Rightarrow$ Now $16\left(2^{n}-1\right)=2^{c+1}\times\left(2^{n}-1\right), c=3$

16.



$A=\int_{-2}^{1}\left[x+7-x^{2}-1\right]dx+\int_{2}^{1}\left(11+3x-x^{2}-1\right)dx$

$=\dfrac{50}{3}, 6A=100$

17. Actual mean $\mu=\dfrac{100(40)-50+40}{100}=39.9$

$\mu=39.9$ incorrect variance

$\therefore (5.1)^{2}=\dfrac{\sum x_i^{2}}{100}-\left(\bar{x}\right)^{2}$

$\therefore \sum x_i^{2}=100\times(40)^{2}+100(5.1)^{2}=162601$

$\sigma^{2}=1617.01-(39.9)^{2}=25$

$\sigma=5$

Then $100(\mu+\sigma)=4490$

18. Let $E_1=$ 3 red, 7 green, $E_2=$4 red, 6 green

$E_3=5$ red, 5 green, $E_4=$ 6 red, 4 green

$E_5=7$ red, 3 green, $A=3$ red, 3 green ball drawn

Then $P(E_1)=P(E_2)=P(E_3)=P(E_4)=P(E_5)=\dfrac{1}{5}$

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$P\left(\frac{A}{E_1}\right)=\frac{1}{6}, P\left(\frac{A}{E_2}\right)=\frac{8}{21}, p\left(\frac{A}{E_3}\right)=\frac{10}{21}, P\left(\frac{A}{E_4}\right)=\frac{8}{21}, P\left(\frac{A}{E_5}\right)=\frac{1}{6}$

Required probability

$$\therefore P\left(\frac{E_3}{A}\right)=\frac{P(E_3).P\left(\frac{A}{E_3}\right)}{P(E_1)P\left(\frac{A}{E_1}\right)+P(E_2).P\left(\frac{A}{E_2}\right)+...P(E_5)P\left(\frac{A}{E_5}\right)}=\frac{10}{33}$$

19. Conceptual

20. $f(x)=1-2x+e^x\int_0^x e^{-t}f(t)dt$

$$e^{-x}f(x)=(1-2x)e^{-x}+\int_0^x e^{-t}f(t)dt$$

$$e^{-x}f(x)-e^{-x}f(x)=-2e^{-x}+(1-2x)e^{-x}(-1)+e^{-x}f(t)$$

$$f(x)-2f(x)=2x-3$$

$$\frac{dy}{dx}-2y=2x-3$$

$$\Rightarrow y\cdot e-2x=\int e^{-2x}(2x-3)dx$$

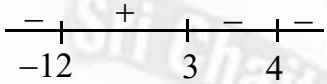
On solving we get

$f(x)=1-x$

$$g(x)=\int_0^x(3-t)^{15}(t-4)^6(t+12)^{17}\,dt$$

$$g'(x)=(3-x)^{15}(x-4)^6(x+12)^{17}$$

$$=-(x-3)^{15}(x-4)^6(x+12)^{17}$$



Local maxima $\Rightarrow q=3$

Local minima $\Rightarrow p=-12$

$$=\frac{|p+q|}{3}=3$$

21. $x+y+z+t=29 \quad ....(1)$

x,y,z,t are integers

$x\geq 1, y\geq 2, z\geq 3, t\geq 0$

$x-1\geq 0, y-2\geq 0, z-3\geq 0, t\geq 0$

$x_1=x-1, x_2=y-2, x_3=z-3$

$x=x_1+1, y=x_2+2, z=x_3+3$

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From (1), we get

$x_1+1+x_2+2+x_3+3+t=29$

$x_1+x_2+x_3+t=23$

Total no of solutions $23+4-1_{C_{4-1}}=26_{C_3}=2600$

22. $ae=\sqrt{10},\frac{a}{e}=\frac{9}{10},a^2=9,e=\frac{\sqrt{10}}{3}$

Now $(ae)^2=a^2+b^2$

$10=9+b^2,b^2=1$

$\ell=L.L.R=\frac{2b^2}{a}=\frac{2(1)}{3}=\frac{2}{3}$

$\therefore 27\left(e^2+\ell\right)=27\left(\frac{10}{9}+\frac{2}{3}\right)=48$

23. $\log_4\log_6\left(3+4x-x^2\right)>0$

$\log_6\left(3+4x-x^2\right)>1$

$3+4x-x^2>6 \qquad x^2-4x+3<0\Rightarrow x\in(1,3) \qquad$ So $a=1,b=3$

$\therefore\int_0^2\left(x^2\right)dx=\int_0^1\left[x^2\right]dx+\int_1^{\sqrt{2}}\left[x^2\right]dx+\int_{\sqrt{2}}^{\sqrt{2}}\left[x^2\right]dx+\int_{\sqrt{3}}^{\sqrt{4}}\left[x^2\right]dx$

$=0+[x]_1^{\sqrt{2}}+2[x]_{\sqrt{2}}^{\sqrt{3}}+3[x]_{\sqrt{3}}^{\sqrt{4}}=5-\sqrt{2}-\sqrt{3} \qquad \frac{p+r}{a}=\frac{5+3}{2}=4$

24. $(2x+3y-1)dx+(4x+6y-5)dy=0$

Let $u=2x+3y-1$ $\frac{dy}{dx}=\frac{1}{3}\left(\frac{dy}{dx}-2\right)$ equation become

$4+(2u-3)\frac{1}{3}\frac{dy}{dx}-2=0$

Use separable variable method

$\alpha=3,\beta=6,\gamma=7 \quad 3\alpha+2\beta+\gamma=9+12+7=28$

25. $\sum_{r=1}^{k}r^3=\frac{k^2(k+1)^2}{4},\cot^{-1}x=\tan^{-1}\left(\frac{1}{x}\right)$

Let $I=\frac{1}{\pi}\sum_{k=1}^{\infty}\tan^{-1}\left(\frac{1}{1+k(k+1)}\right) \qquad =\frac{1}{\pi}\sum_{k=1}^{\infty}\tan^{-1}(k+1)-\tan^{-1}k=\frac{1}{\pi}\times\frac{\pi}{4}=\frac{1}{4}$

$I=3\sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n\Rightarrow G.P \qquad =3\times\frac{1/4}{1-\frac{1}{4}}=3\times\frac{1}{3}=1$

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PHYSICS

26. A) Plancks constant $h=\frac{E}{\vartheta}\left(E=h\vartheta\right) \quad =\frac{ML^2T^{-2}}{T^{-1}}=ML^2T^{-1}$

B) Momentum (P) $=mv=M^1L^1T^{-1}$

C) Moment of Intertia $I=MR^2=M^1L^2$

D) Torque $\vec{T}=\vec{r}\times\vec{F}=M^1L^2T^{-2}$

27. $mg=\left(mg\right)' \Rightarrow d_bv_bg=d_lv_lg$

$\rho_bAH_b=\rho_lAH_l \quad 800\times10=1200\times H_l \qquad \Rightarrow H_l=\frac{80}{12}=6.3$

28. Area $A=\frac{\pi h^2}{\mu^2-1}.\quad \pi r^2=\frac{\pi h^2}{\mu^2-1}\Rightarrow r=\frac{h}{\sqrt{\mu^2-1}}=\frac{2\sqrt{7}}{\sqrt{\frac{16}{9}-1}}$

$$r=\frac{2\sqrt{7}}{\sqrt{\frac{7}{9}}}=2\times3=6m$$

29. $\Delta K=\Delta K_1+\Delta K_2=0.2+0.3=0.5$

$K_{eq}=K_1+K_2=10+15=25$

Percentage error $=\frac{\Delta K}{K}\times100 \qquad =\frac{0.5}{25}\times100=2\%$

30. Uniform angular velocity implies a constant direction of rotation and speed of rotation constant, angular acceleration is rate of charge of angular velocity

31. $\varepsilon=Blv \qquad =0.1\times1\times2=20\times10^{-2}$

$I=\frac{\varepsilon}{R} \qquad =\frac{20\times10^{-2}}{4}=5\times10^{-2}$

$F=BIl \qquad =0.1\times5\times10^{-2}\times1 \quad =5\times10^{-3}\,N$

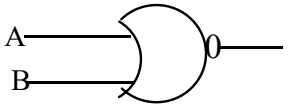
32. Range $R=\frac{u^2\sin2\theta}{g} \qquad R\propto\sin2\theta \qquad \frac{R_1}{R_2}=\frac{\sin\left(2\theta_1\right)}{\sin\left(2\theta_2\right)}$

$\frac{40}{R_2}=\frac{\sin30^\circ}{\sin90^\circ} \qquad R_2=80m$

33. With increase in temperature, the value of unknown resistance will increases. then null point shifting towards lower resistance side.

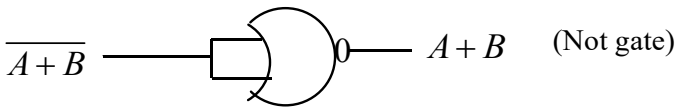
34. $s=\frac{v^2}{2\mu_kg}=\frac{4.9\times4.9}{2\times0.5\times9.5}=\frac{4.9}{2}=2.45m$

35.



$\overline{A+B}$ (Nor gate)

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36. $U_i = \dfrac{-GM_e m}{\mathrm{Re}} \qquad U_f = \dfrac{-GM_e m}{\mathrm{R}_e + 3R_e} = \dfrac{-GM_e m}{4\mathrm{R}_e}$

Increases in internal energy $\Delta U = U_f - U_i = \dfrac{3}{4} mg\,\mathrm{R}_e$

37. $Z = \sqrt{R^2 + X^2} = \sqrt{3^2 + 4^2} = 5\Omega; \cos\phi = \dfrac{R}{z} = \dfrac{3}{5} = 0.6$

38. Area under P-V graph

$W = \dfrac{1}{2} \times base \times height \qquad = \dfrac{1}{2} \times 1 \times 100 = 50J$

39. $\tan\theta = \dfrac{i}{v} = \dfrac{1}{R},\ R = \dfrac{1}{\tan\theta} = \dfrac{1}{\sqrt{3}}$

40. $g = \dfrac{GM}{R^2} \qquad g' = \dfrac{G2M}{(4R)^2} = \dfrac{2GM}{16R^2} = \dfrac{g}{8}$

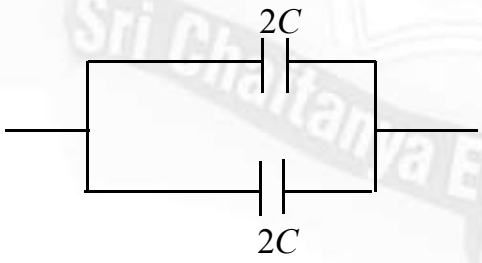
Assertion false $T = 2\pi\sqrt{\dfrac{\ell}{g}}$

Time period is independent of the mass of pendulum

41. $a = \dfrac{F}{m} = \dfrac{Ee}{m} = \dfrac{10^3 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.8 \times 10^{14}\ ms^{-2}$

$s = \dfrac{v^2 - u^2}{2a} = \dfrac{0 - \left(5 \times 10^6\right)^2}{-2 \times 1.8 \times 10^{+14}} = 0.07\,m$

42. The circuit can be reduced to



$Ceq = 2C + 2C = 4C$

43. The photoelectron will have a range of kinetic energies, because after absorbing the photon, electron's in this metals collide with other atoms before being ejected out of metal.

44. $A = \pi R^2; R = \sqrt{\dfrac{A}{\pi}}$

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$B=\frac{\mu_0 i}{2R}\Rightarrow i=\frac{2BR}{\mu_0};M=iA=\left(\frac{2BR}{\mu_0}\right)A \qquad \text{A}M=\frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$

45. $P=\frac{nE}{t} \qquad \frac{n}{t}=\frac{P}{E}=\frac{10^7}{200\times1.6\times10^{-13}}=3.125\times10^{17}\ atoms$

46. $\Delta T=\frac{1}{2}T\propto\Delta\theta \qquad =\frac{1}{2}\times2\times\left(1.2\times10^{-5}\right)(40-20)$

$=1.2\times10^{-5}\times20=2.4\times10^{-4}\,s$

New time period $T'=\Delta T+T \qquad =\left(2.4\times10^{-4}+2\right)=2.00024\,s$

Time lost in one week $\Delta t=\left(\frac{\Delta T}{T}\right)t=\frac{2.4\times10^{-4}}{(2.00024)}(7\times24\times3600)=72.567=73$

47. $\frac{1}{f_A}=(\mu_A-1)\left(\frac{1}{R_1}-\left(\frac{1}{-R_2}\right)\right)$

$\frac{1}{f_B}=(\mu_B-1)\left(\frac{1}{\infty}+\frac{1}{R_2}\right)$

$|P_A|=|P_B|=0.5\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=\frac{0.6}{R_2} \qquad \frac{0.5}{R_1}=\frac{0.1}{R_2} \qquad \frac{R_1}{R_2}=\frac{5}{1}$

48. In the second excited state $(n=3)$ of $E_H=\frac{-13.6}{9}=-1.51eV$ H-atom

I the first excited state $(n=2)$ of Li^{2+} ion

$E_{Li}=\frac{-13.6\times(3)^2}{(2)^2}=-30.6eV$

Ratio of debroglie wavelength is $\frac{\lambda_H}{\lambda_{Li}}=\sqrt{\frac{E_{Li}}{E_H}}=\sqrt{\frac{30.6}{1.51}}=4.5=\frac{9}{2}$

49. $I=2\times\frac{7}{5}MR^2 \qquad =\frac{14}{5}\times4\times100\times10^{-4}$

$=56\times20\times10^{-4} \qquad =1120\times10^{-4} \qquad =112\times10^{-3}\,kg\,m^2 \quad x=112$

50. $C=f\lambda \qquad \lambda_{air}=\frac{c}{f}=\frac{3\times10^8}{6\times10^{14}}=5\times10^{-7}\,m$

$\lambda_{water}=\frac{\lambda_{air}}{\mu_{water}}=\frac{5\times10^{-7}}{4/3} \qquad \lambda_w=3.75\times10^{-7}\,m \;\; B=\frac{\lambda_w D}{d}\Rightarrow D=1m$

$\Rightarrow d=0.2\ mm=0.2\times10^{-3}m$

$\beta=\frac{3.75\times10^{-7}\times1}{0.2\times10^{-3}}m \qquad \beta=187\times10^{-5}=1.87\times10^{-3}$

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CHEMISTRY

51. $n=3,\ell=3$ is not possible as $\ell=0,1,...(n-1)$

52. Vapour pressure of component I is less than component 2

53. 1) The correct order of size is $Mg>Al>Mg^{+2}>Al^{+3}$ atomic size depends mainly upon effective and shell numbers
2) Generally an moving down the group electron affinity decreases and an moving across the period electron affinity increases.
$\therefore$ In periodic false Cl has maximum electron affinity halogens has high EA than chalogas
$Cl>Br>S>O$

54. Assertion (A): PH_3 is more basic than NH_3.
Reason (R): Electro negativity of nitrogen is greater than phosphorus.

55. 1st ionisation enthalpy order
F > N > O > Be > B

56. $\wedge_{eq}=\dfrac{K\times1000}{N}\Rightarrow K=\dfrac{\wedge_{eq}\times N}{1000}=\dfrac{100\times0.01}{1000}=0.001S\,cm^{-1}$

$G=\dfrac{K}{G^*}=\dfrac{0.01}{0.25}=0.004\,ohm^{-1}$

$R=\dfrac{1}{G}=\dfrac{1}{0.04}=250\ ohm$

$Current(i)=\dfrac{V}{R}=\dfrac{5}{250}=0.02A$

57. $K_2=\dfrac{1}{t}\ln\dfrac{100}{4};K_1=\dfrac{1}{t}\ln\dfrac{100}{50}\qquad\dfrac{K_2}{K_1}=\dfrac{\ln25}{\ln2}=4.65$

58. $NaCN>NaCl>NH_4Cl>HCl$
$NaCN$ →Basic salt→ highesh pH
HCl →strong acid→ lowest pH
$NaCl$ →Neutral salt→ pH = 7
NH_4Cl →acidic salt→ pH < 7

59. $NaClO_{3(s)}+Fe_{(s)}\longrightarrow O_{2(g)}+NaCl_{(s)}+FeO_{(s)}$
Moles of $NaClO_3=moles\ of\ O_2$

$moles\ of\ O_2=\dfrac{PV}{RT}=\dfrac{1\times246}{0.082\times300}=10\,moles$

60. Work done $=-2.303nRT\log\dfrac{V_2}{V_1}$

$w=-2.303\times2\times8.314\times10^{-3}\times300\log\dfrac{40}{4}=-11.488KJ$

61. In $C_2H_4\Rightarrow$ C = C bond is not equal to C-H bond.

62. In glycogen branching occurs due to C_1-C_6 of glycosidic linkage of $\alpha-D-$glucose units .

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63. Bond order $= \frac{N_b - N_a}{2}$ For $N_2 = \frac{10-4}{2} = 3$

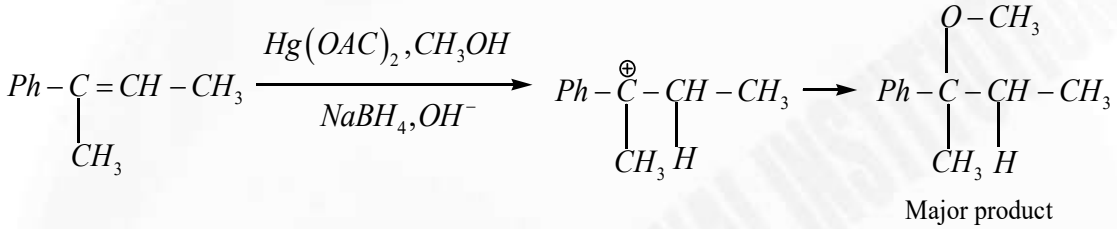
For $O_2^- = \frac{10-7}{2} = 1.5$ For $O_2^{-2} = \frac{10-8}{2} = 1$

64. Cr^{+3} has 3 unpaired electrons so it's $\mu = \sqrt{3(3+2)} = 3.83BM$.

$\left[Cr(H_2O)_4Cl_2\right]Cl.2H_2O$ exhibits geometrical isomerism.

65. CFSF of $\Delta_t = \frac{4}{9}\Delta_o = \frac{4}{9} \times 18000\,cm^{-1} = 4 \times 2000\,cm^{-1} = 8000cm^{-1}$

66.



Major product

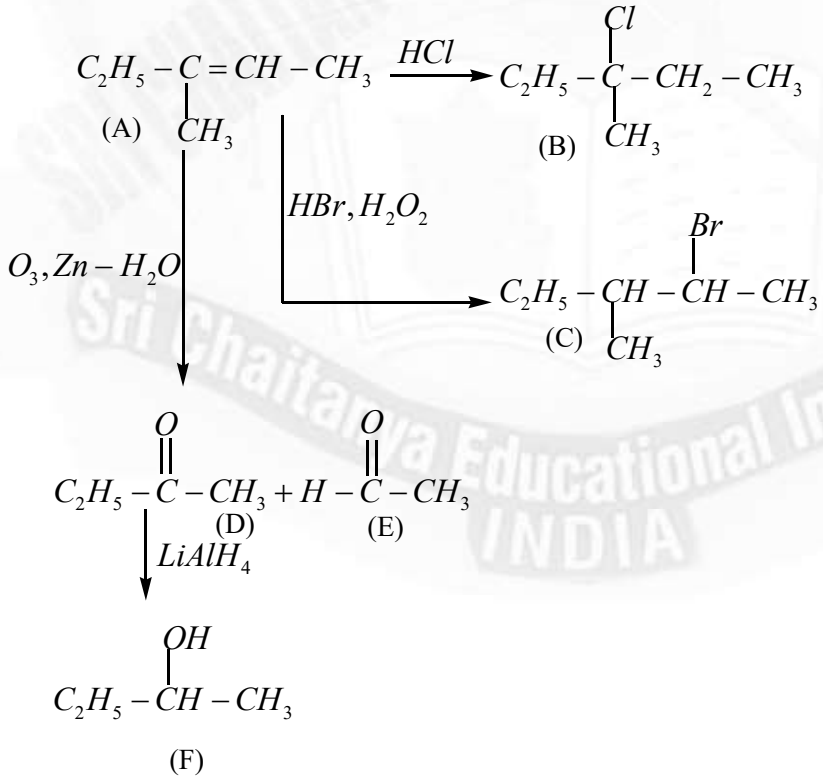
67. $2Cu^{+2} + 8CN^- \rightarrow 2\left[Cu(CN)_4\right]^{-3} + (CN)_2$

Cu^{+1} is more stable so it doesn't reacts with H_2S.

68. I^- is large in size in $C_6H_5O^-$ resonance is present.

69. Both aldehydes and terminal alkynes gives positive Tollen's test. Both aldehyde and ketones gives positive 2,4-DNP test. So they can't distinguished by 2,4 – DNP.

70.



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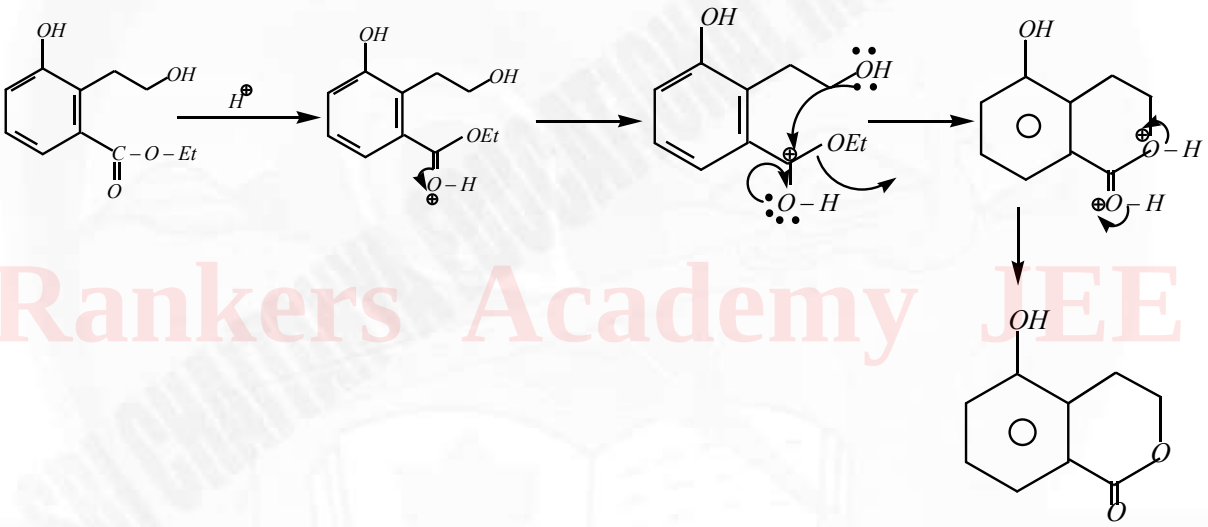
71. $T = \dfrac{\Delta H_{vap}}{\Delta S_{vap}} = \dfrac{30000}{75} = 400K$

72. n-factor of acid = basicity (n)
Equation of acid = equation of base
$\dfrac{2}{600} \times n = \dfrac{1}{6} \times \dfrac{100}{1000} \times 1$
$n = 5$

73. $R_f = \dfrac{Distance\ moved\ by\ the\ substance\ from\ the\ base\ line}{Distance\ moved\ by\ the\ solvent\ from\ the\ base\ line}$

$R_f = \frac{3.5}{7} = 0.5\ or\ 5 \times 10^{-1}$

74.



No. of atom $x = 3$
No. of sp^2 'C' atoms $y = 7$
$\dfrac{x+y}{2} = \dfrac{3+7}{2} = 5$

75. $\%H = \dfrac{2}{18} \times \dfrac{0.15}{0.25} \times 100 = 6.6\ \%$
$\%C = \dfrac{12}{44} \times \dfrac{0.1830}{0.25} \times 100 = 19.76\ \%$
$0\% = $ Subtraction method
$100 - (\%H + \%C) = 73.64\% \approx 74\%$

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